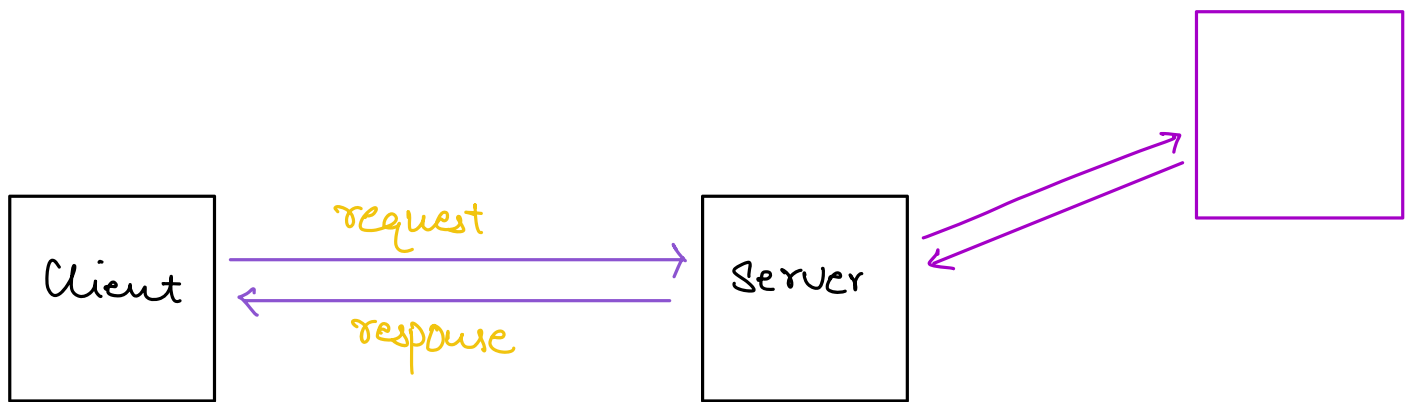


API

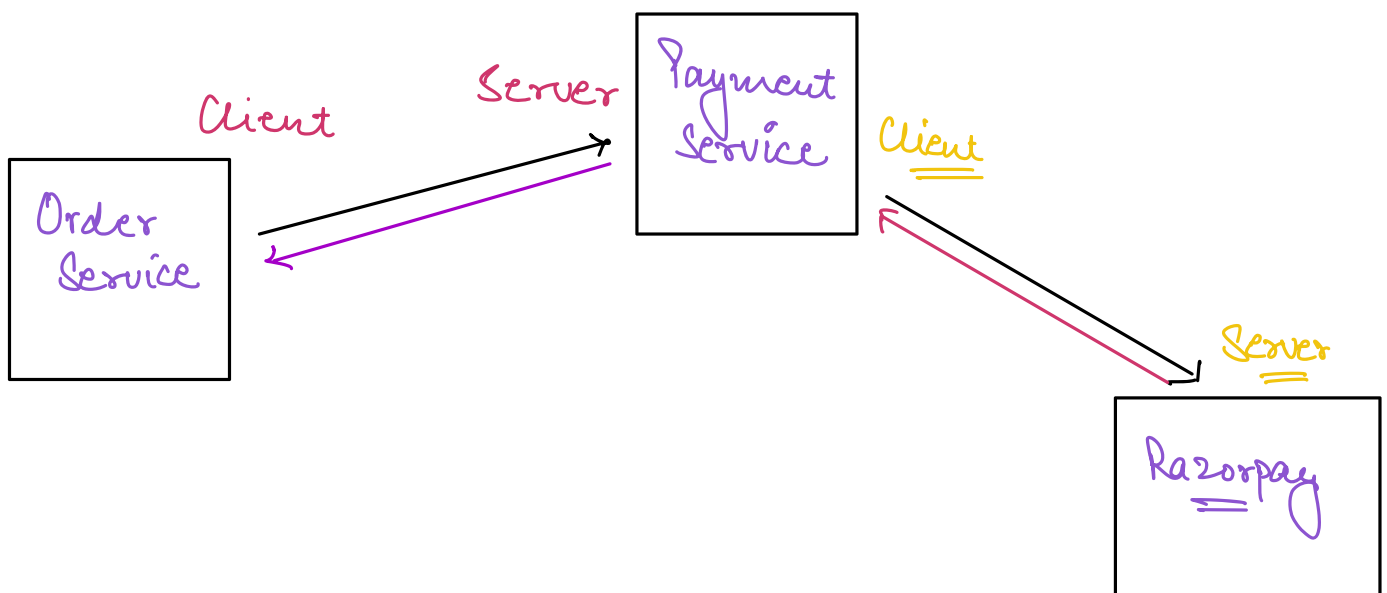
↳ Application Programming Interface.

→ It is a contract between two software systems that allows them to communicate with each other.

Request - Response Cycle.



Client - Server Architecture.



HTTP Methods.

- GET : Read | fetch
- POST : creating
- PUT | PATCH : Update
- DELETE : Delete.

GET /orders ⇒ fetch all the orders

GET /orders/123 ⇒ fetch the order with id = 123.

HTTP Status Codes.

- 200 : OK | Request Successful.
- 201 : New resource create successfully.
- 404 : Not found
- 401 : Unauthorize

2xx	Success
4xx	Client's Mistake
5xx	Server's Mistake

JSON.

→ Java Script Object Notation.

→ Most common data format used for communication b/w software systems.

```
{  
  "id": _____,  
  "name": _____,  
  "age": _____,  
  "_____": _____,  
}
```

function calling

→ function calling means giving an AI model a list of available tools/functions and allowing it to decide which tool to call.

The Function Calling Flow

Step 1: User asks a question.

Step 2: AI model checks whether it can answer directly.

Step 3: If external data/action is needed, model selects a tool.

Step 4: Model generates structured arguments for that tool.

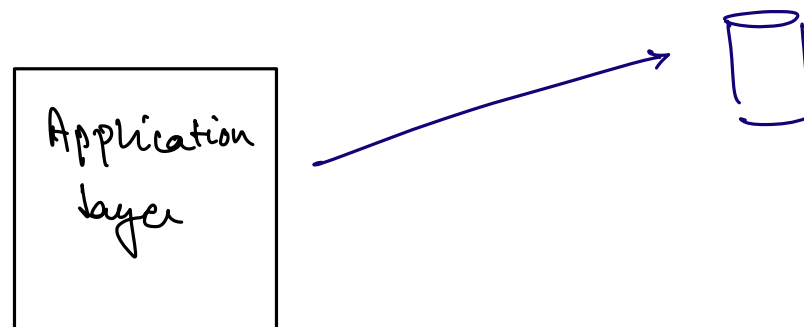
Step 5: Application validates the arguments.

Step 6: Application executes the actual Python function/API call.

Step 7: Tool returns structured data.

Step 8: Application sends tool result back to the model.

Step 9: Model generates final user-friendly response.



Database Drivers

ORM : Object Relational Mapping

Class User :

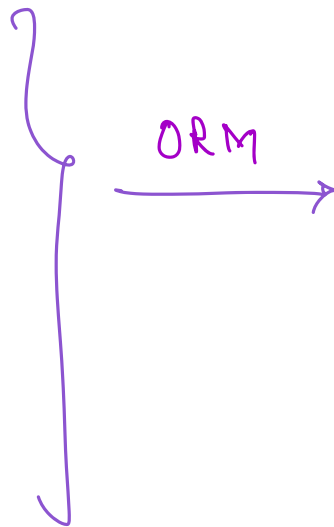
id : —

name : —

age : —

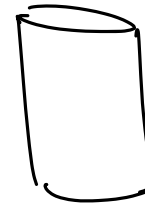
college : —

marks : —



user

id	name	age	college	marks



SQLAlchemy + ORM

SQLAlchemy.

→ Python library that helps us to interact with Database

- 1) SQLAlchemy Core
- 2) SQLAlchemy ORM

Python Class → Database Table

